SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships.* (BL1, BL2)

Activity Tool : Short Answer Text(High)
Assessment Tool Weightage: 30%

Dr

QUESTIONS		Total Marks: 50	
Q.No	**Question**	**Bloom's Level**	**Marks**
1	UPS Product Tracking Information System UPS prides itself on having up-to-date information on the processing and current location of each shipped item. To do this, UPS relies on a company-wide information system. Shipped items are the heart of the UPS product tracking information system. Shipped items can be characterized by item number (unique), weight, dimensions, insurance amount, destination, and final delivery date. Shipped items are received into the UPS system at a single retail center. Retail centers are characterized by their type, uniqueID, and address. Shipped items make their way to their destination via one or more standard UPS transportation events (i.e., flights, truck deliveries). These transportation events are characterized by a unique scheduleNumber, a type (e.g, flight, truck), and a deliveryRoute. Create an Entity Relationship diagram that captures this	BL1 – Remember	5

	information about the UPS system.		
2	Course Management System Consider a university database for the scheduling of classrooms for final exams. This database could be modeled as the single entity set exam, with attributes course-name, section-number, room-number, and time. Alternatively, one or more additional entity sets could be defined, along with relationship sets to replace some of the attributes of the exam entity set, as • course with attributes name, department, and c-number • section with attributes s-number and enrollment, and dependent as a weak entity set on course • room with attributes r-number, capacity, and building Show an E-R diagram illustrating the use of all three additional entity sets listed.	BL2 – Understand	5
3	University Registrar's Office Management System A university registrar's office maintains data about the following entities: (a) courses, including number, title, credits, syllabus, and prerequisites; (b) course offerings, including course number, year, semester, section number, instructor(s), timings, and classroom; (c) students, including student-id, name, and program; (d) instructors, including identification number, name, department, and title. (e) Further, the enrollment of students in courses and grades awarded to students in each course they are enrolled for must be appropriately modeled. Construct an E-R diagram for the registrar's office.	BL2 – Understand	5

4	Hospital Management System Construct an E-R diagram for a hospital with a set of patients and a set of medical doctors. Associate with each patient a log of the various tests and examinations conducted. Construct appropriate tables for the above ER Diagram : Patient(SS#, name, insurance) Physician (name, specialization) Test-log(SS#, test-name, date, time) Doctor-patient (physician-name, SS#) Patient-history(SS#, test-name, date)	BL2 – Understand	5
5	Company Project Management System Construct an ER Diagram for Company having following details : Company organized into DEPARTMENT. Each department has unique name and a particular employee who manages the department. Start date for the manager is recorded. Department may have several locations. A department controls a number of PROJECT. Projects have a unique name, number and a single location. Company's EMPLOYEE name, ssno, address, salary, sex and birth date are recorded. An employee is assigned to one department, but may work for several projects (not necessarily controlled by dept). Number of hours/week an employee works on each project is recorded; The immediate supervisor for the employee. Employee's DEPENDENT are tracked for health insurance purposes (dependentname, birthdate, relationship to employee).	BL2 – Understand	5
6	Explain schema and instance with examples. Distinguish between logical schema, physical schema, and view schema.	BL1 – Remember	5
7	Identify the entities, attributes, and relationships for a Bank Management System and construct its ER diagram.	BL2 – Understand	5

8	Explain the Extended ER (EER) model. Describe the concepts of generalization, specialization, and aggregation.	BL2 – Understand	5
9	Design an EER diagram for a Hospital Management System incorporating inheritance and weak entities.	BL2 – Understand	5
10	Explain the role of a DBA (Database Administrator) and the responsibilities associated with the position.	BL1 – Remember	5

Code of Conduct:

I R Harika(192572140)____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

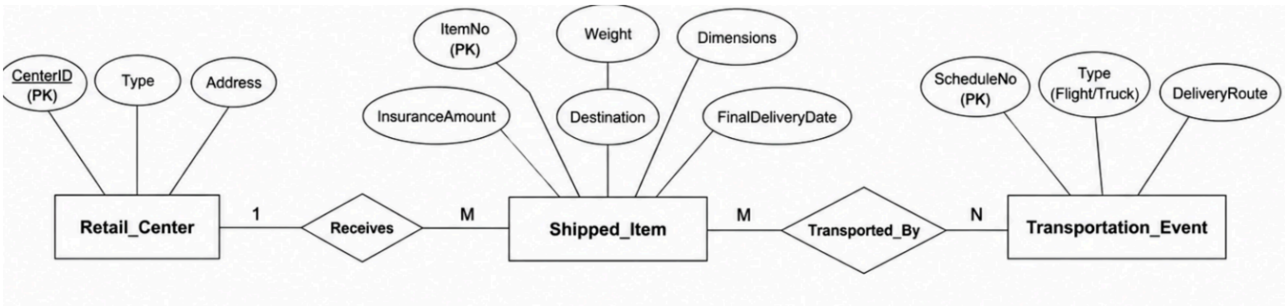
Signature of the student:

RUBRICS FOR CALCULATION:

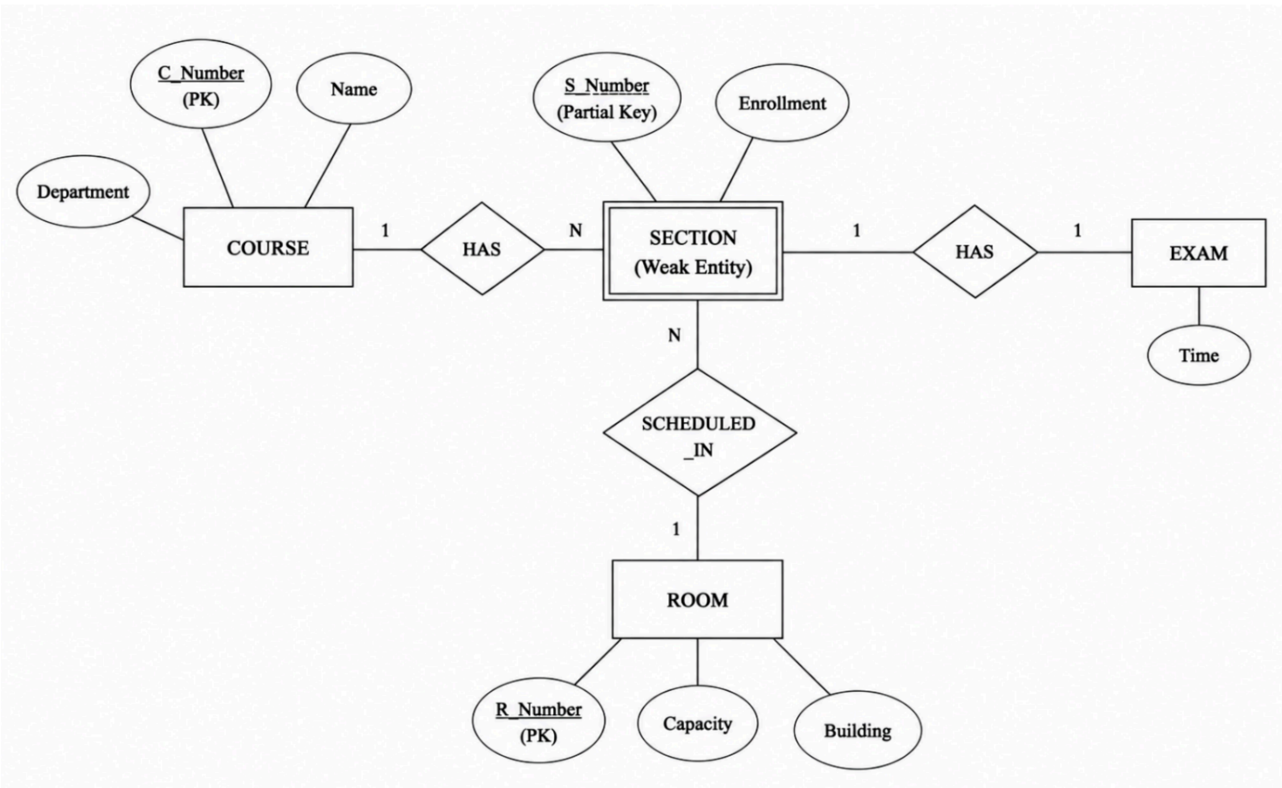
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Concept Identification	30%	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understandin g
Linkages	25%	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer

Organization	20%	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Integration of Literature	15%	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity	10%	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

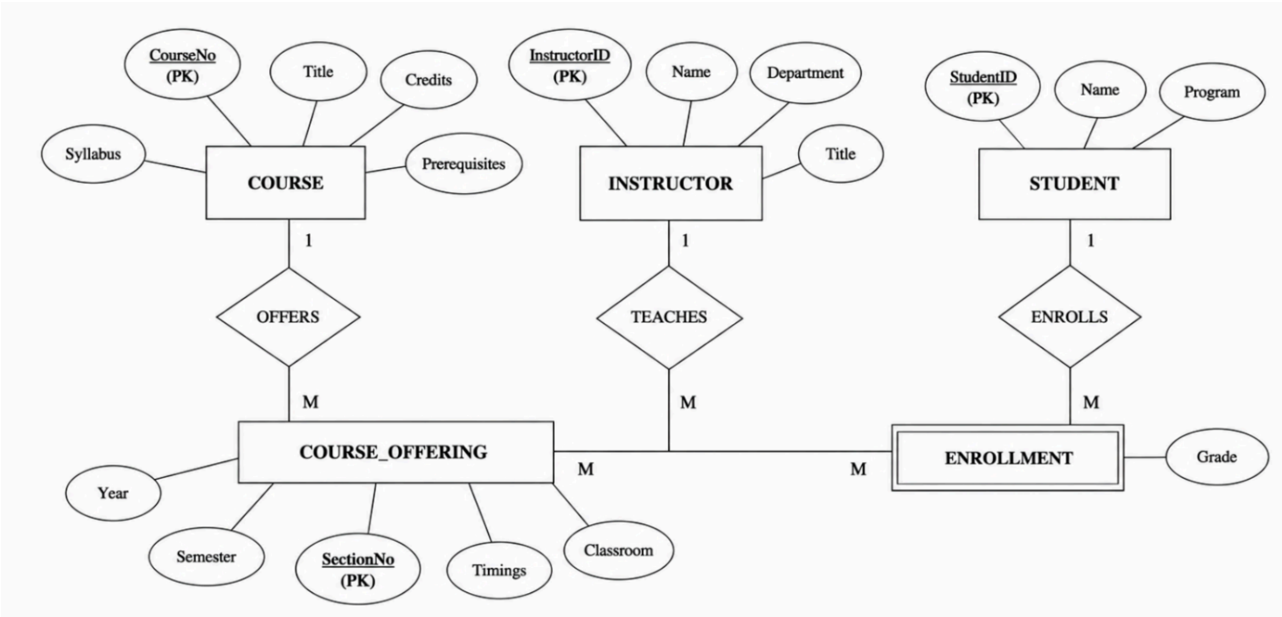
1.



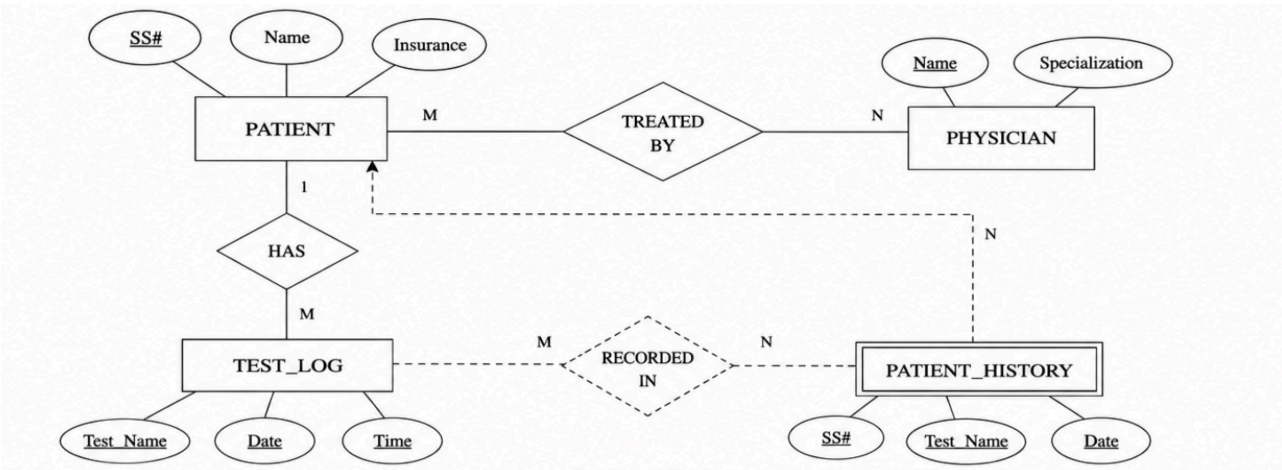
2.



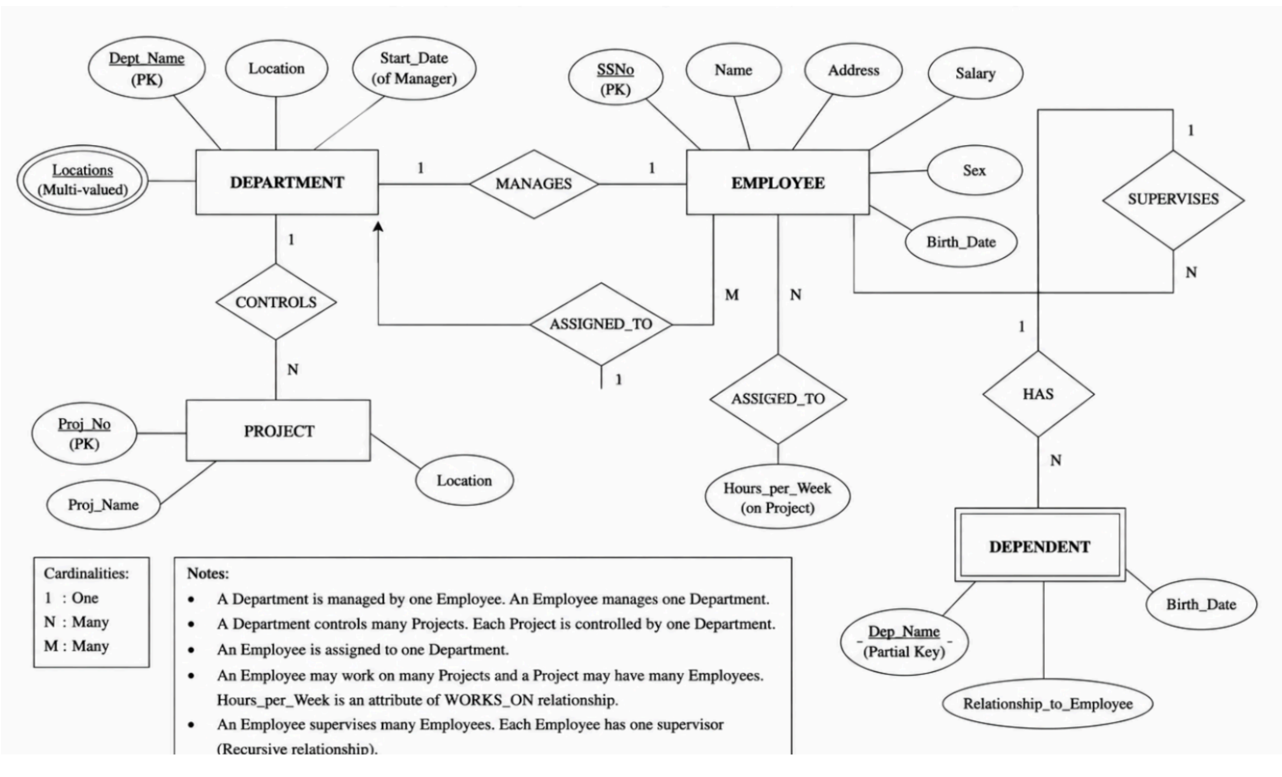
3.


CourseNo (PK)
Title
Credits
Syllabus
Prerequisites
COURSE
InstructorID (PK)
Name
Department
Title
INSTRUCTOR
StudentID (PK)
Name
Program
STUDENT
1
OFFERS
TEACHES
ENROLLS
M
COURSE_OFFERING
ENROLLMENT
Grade
Year
Semester
SectionNo (PK)
Timings
Classroom

4.


SS#
Name
Insurance
PATIENT
M
TREATED BY
N
Name
Specialization
PHYSICIAN
1
HAS
M
TEST_LOG
RECORDED IN
PATIENT_HISTORY
Test_Name
Date
Time
SS#
Test_Name
Date

5.


Dept_Name (PK)
Location
Start_Date (of Manager)
Locations (Multi-valued)
DEPARTMENT
1
MANAGES
EMPLOYEE
SSNo (PK)
Name
Address
Salary
Sex
Birth_Date
SUPERVISES
N
CONTROLS
ASSIGNED_TO
M
ASSIGED_TO
HAS
Hours_per_Week (on Project)
PROJECT
Proj_No (PK)
Proj_Name
Location
DEPENDENT
Dep_Name (Partial Key)
Birth_Date
Relationship_to_Employee
Cardinalities:
1 : One
N : Many
M : Many
Notes:
A Department is managed by one Employee. An Employee manages one Department.
A Department controls many Projects. Each Project is controlled by one Department.
An Employee is assigned to one Department.
An Employee may work on many Projects and a Project may have many Employees. Hours_per_Week is an attribute of WORKS_ON relationship.
An Employee supervises many Employees. Each Employee has one supervisor (Recursive relationship).

6. **Schema**
A **schema** is the overall structure or blueprint of a database. It defines how data is organized, including tables, attributes, relationships, and constraints. A schema changes very rarely.
Example:
STUDENT(Student_ID, Name, Department, Age)

Instance
An **instance** is the actual data stored in the database at a particular point in time. It changes whenever records are inserted, updated, or deleted.
Example:

Student_ID	Name	Department	Age
101	Rahul	CSE	20
102	Priya	IT	19

Difference between Schema and Instance

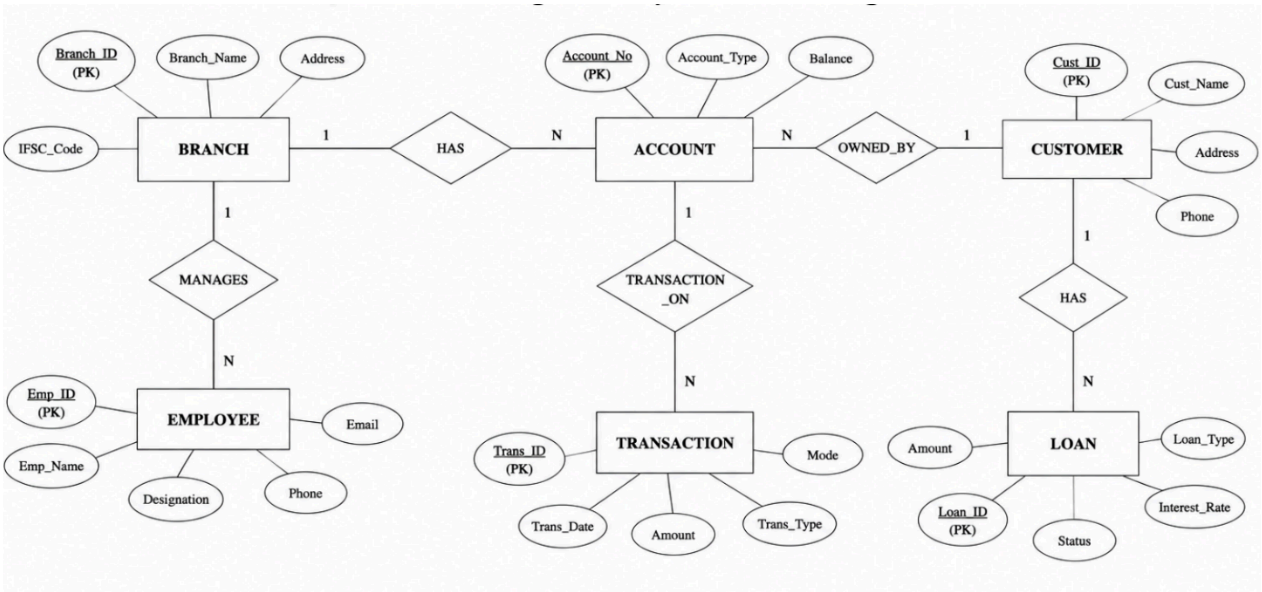
Schema	Instance
Blueprint of the database	Actual data stored in the database
Changes rarely	Changes frequently
Defines structure	Contains current data

Types of Schema
1. Logical Schema
- Defines the logical structure of the database.
- Specifies tables, attributes, relationships, and constraints.
- Independent of physical storage.

2. Physical Schema
- Describes how data is physically stored on disk.
- Includes storage structures, indexing, and file organization.
- Managed mainly by the DBA.

3. View Schema (External Schema)
- Represents the part of the database visible to a particular user.
- Provides security by hiding unnecessary data.
- Different users can have different views.

7.



8. **Extended Entity Relationship (EER) Model**
The **Extended Entity Relationship (EER) Model** is an extension of the basic ER model. It provides advanced features such as **generalization, specialization, inheritance, and aggregation**, making it suitable for modeling complex real-world databases.

1. Generalization

- Generalization is the process of **combining two or more lower-level entities into a higher-level entity** based on their common attributes.
- It follows a **bottom-up approach**.
- Common attributes are placed in the superclass.

Example:

```
Car     Bike
  \      /
   \    /
   Vehicle
```

2. Specialization

- Specialization is the process of **dividing a higher-level entity into lower-level entities** based on distinguishing characteristics.
- It follows a **top-down approach**.
- Subclasses inherit all attributes of the superclass.

Example:

```
      Employee
     /        \
 Manager    Engineer
```

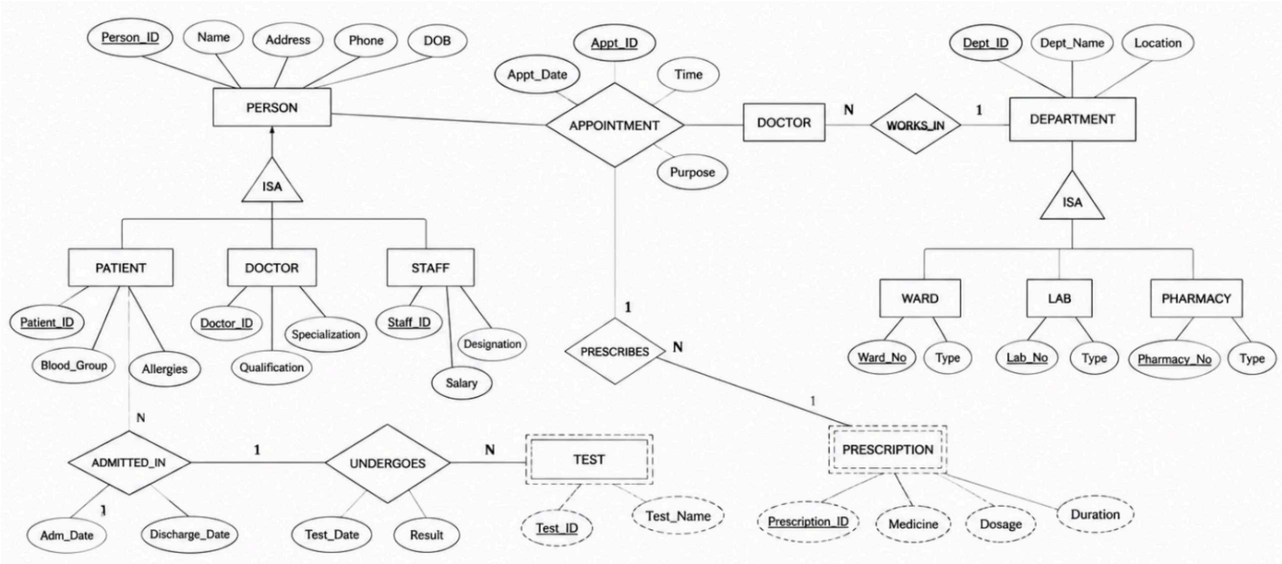
3. Aggregation

- Aggregation is used when a **relationship itself participates in another relationship**.
- It treats a relationship as a higher-level entity.

Example:

```
Employee ---- Works_On ---- Project
                   |
             Supervised_By
                   |
               Manager
```

9.



10. **Database Administrator (DBA)**

A **Database Administrator (DBA)** is a person responsible for installing, managing, securing, and maintaining a database. The DBA ensures that the database is available, reliable, secure, and performs efficiently.

Responsibilities of a DBA

1. **Database Installation and Configuration**
   - Installs and configures the DBMS software.
2. **Database Design**
   - Creates and maintains database structures, tables, and schemas.
3. **Security Management**
   - Creates user accounts, assigns permissions, and protects data from unauthorized access.
4. **Backup and Recovery**
   - Performs regular backups and restores data in case of system failure.
5. **Performance Monitoring**
   - Monitors database performance and optimizes queries for better speed.
6. **Data Integrity**
   - Ensures data accuracy, consistency, and integrity using constraints and validation.
7. **Storage Management**
   - Manages database storage space and allocates memory efficiently.
8. **Database Maintenance**
   - Updates the database, fixes errors, and applies software patches.
9. **Disaster Recovery**
   - Develops recovery plans to prevent data loss during hardware or software failures.
10. **User Support**
    - Assists users in resolving database-related issues and ensures smooth database operations.

Functions of a DBA

- Maintains database security.
- Ensures high availability of data.
- Improves database performance.
- Performs backup and recovery.
- Maintains data consistency and reliability.